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CORE STACK

TypeScriptNode.js

PostgreSQLRedisRabbitMQ

Next.jsReactExpress

Drizzle ORM

ENGINEERING

System Architecture

ConcurrencyRate Limiting

ObservabilityLoad Testing

Distributed State

ANALYTICS

PostHogTally Webhooks

Funnel Analytics

TESTING & TOOLS

VitestGitGitHub

Cloudflare Pages

EDUCATION

Reduzer Training Institute

Software Engineering Bootcamp

Sep 2025 - Sep 2026

St. Augustine's Mirogi Boys High School

KCSE

2015 - 2018

CERTIFICATIONS

- Software Engineering Certificate · Reduzer Training Institute · 2026
- CyberOps Associate · Cisco Networking Academy · 2024
- Digital Marketing Certificate · Aleph Digital Academy · 2023

VINCENT OOKO OCHIENG

SOFTWARE ENGINEER · FULL-STACK & BACKEND SYSTEMS

01. PROFILE

Software Engineer focused on **full-stack development, backend systems, and production-minded engineering**. Builds across the stack, from user-facing applications and APIs to databases, asynchronous systems, analytics, and deployment. Combines **systems thinking with product and data awareness**, with a strong focus on reliability, performance, observability, testing, and measurable outcomes.

02. PROFESSIONAL EXPERIENCE

Reduzer Training Institute · Reduzer School · Software Engineering Trainee Sep 2025 - Present

- Built the PostHog analytics integration for the Reduzer School landing page and the Tally webhook pipeline that sends applicant form submissions into PostHog to enrich behavioral data.
- Led analysis of website and applicant data, including pageviews, traffic sources, and sign-up conversion events, to understand acquisition, engagement, and where applicants dropped off in the flow.
- Diagnosed a production issue where analytics events weren't firing; traced it to missing NEXT_PUBLIC_* environment variables in the Cloudflare Pages build configuration and restored event capture.
- Contributed to the landing page build alongside deployment and QA work, using TypeScript, Next.js, React, PostHog, and Tally.

Self-Directed Software Development · Independent Developer Before Sep 2025

- Built a foundation in software engineering through self-directed study and hands-on development across JavaScript, React, SQL, databases, APIs, and backend systems.
- Learned by turning documentation and concepts into working software, deliberately experimenting, breaking implementations, debugging failures, and iterating toward more reliable designs.

03. ENGINEERING PROJECTS

01. LOG INGESTION ENGINE Node.js · RabbitMQ · PostgreSQL · Redis

- Built an asynchronous log ingestion pipeline designed around backpressure, failure isolation, and reliable delivery.
- Implemented a 5-second RabbitMQ publish timeout, secondary fallback path, exponential-backoff retries, and rotating JSON dead-letter storage instead of silently dropping failed events.
- Prevented overlapping flush operations with per-destination concurrency guards; load-tested 5,000 requests with 0 failures at 1,385 req/s, P95 1,101ms.
- Built a live metrics dashboard (auto-refreshing every 5s) surfacing throughput, error rate, and queue backlog in real time.

02. RATE LIMITER MIDDLEWARE Node.js · Express · Redis · Lua · Vitest

- Built reusable middleware implementing fixed-window and token-bucket algorithms with configurable caller identification, policies, and HTTP rate-limit signals.
- Used atomic Redis Lua scripts to maintain distributed state safely across concurrent requests and multiple application instances.

03. SQL PERFORMANCE ANALYZER PostgreSQL · SQL

- Built a performance analyzer against a live PostgreSQL database to execute queries and inspect query plans and performance characteristics.
- Created the tool to investigate query efficiency and database bottlenecks, developing practical experience with database performance analysis.

04. JOB MONITORING SYSTEM React · Vite · WebSockets · Node.js · Drizzle ORM

- Built a multi-process system (producer, workers, API server) with a live React/Vite dashboard streaming job status in real time over WebSockets.
- Modeled job and queue state with Drizzle ORM against PostgreSQL, keeping the dashboard's live view in sync with actual worker activity.